# KALAIMANI R

## BACKEND / API DEVELOPER | TECHNICAL LEAD

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Portfolio: kalaimani1502.github.io/kalai_portfolio

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## PROFESSIONAL SUMMARY

Backend / API Developer with **6 years** of experience at **Compunet Connections**, designing and building scalable APIs and web applications across **PHP (Laravel)** and **Python (FastAPI)**. 3+ years as long-term technical owner of a large-scale Australia-based real estate platform — REST APIs, a **custom e-signature engine built in-house**, **WebSocket-based real-time communication**, and currently an **AI-powered chat assistant** (FastAPI, RAG, LLM integration with Ollama/Gemini) — plus Python APIs for transport and event-management applications. Comfortable owning backend architecture end-to-end: API design, database work (PostgreSQL/MySQL), AWS infrastructure (EC2/S3/RDS), client communication, and **technically guiding both frontend and backend developers** through integration and delivery.

## CORE SKILLS

**Leadership & Coordination:** Technical Team Leadership, Developer Guidance & Mentoring, Code Review, Client Communication, Requirement Analysis, Agile/Scrum

**Languages & Frameworks:** PHP (Laravel, Lumen, Slim), Python (FastAPI)

**AI / LLM:** RAG (Retrieval-Augmented Generation), LLM integration (Ollama, Gemini), FAISS and OpenAI API (project exposure) — AI chat assistant in development

**API & Real-Time:** REST APIs, WebSockets, API Architecture, Authentication & Authorization, Swagger/OpenAPI, Postman

**Databases:** PostgreSQL, MySQL, Database Design, Query Optimization, Data Modeling

**Integrations:** Google Maps API, MSG91 SMS with DLT configuration, Moxo (webhooks & client communication)

**Cloud & DevOps:** AWS (EC2, S3, RDS), Jenkins, Git/Bitbucket, Linux

**Frontend & AI-Assisted Dev:** React.js, React Native, HTML/CSS (guidance level) | Claude, ChatGPT, DeepSeek, Qodo — structured workflow with manual review

## PROFESSIONAL EXPERIENCE

**Compunet Connections** | *Backend / API Developer* | *6 Years (Current)*

**API / Backend Developer** | *Project: Australia-Based Real Estate Platform • Current, 3+ Years*

- Designed and maintained scalable **REST APIs** (PHP/Laravel) powering complex real-estate workflows — bidding, legal processes, document management, and internal communication.
- Built a **custom e-signature solution** in-house (no third-party service), owning backend logic and API design while guiding the React.js frontend build.
- Implemented **WebSocket-based real-time communication** for live updates and coordinated frontend/backend integration end-to-end.
- Integrated **Moxo** (client communication & file-management platform) for webhook-based notifications and in-app client communication.
- Worked with **AWS EC2, S3, and RDS** for application hosting, file storage, database, and server-side configuration/troubleshooting.
- Worked directly with international clients to gather requirements and translate business needs into technical solutions.
- Technically guided frontend and backend developers on API design, integration, and debugging; led code reviews and production troubleshooting.
- Currently building an **AI-powered chat assistant** for the platform using **FastAPI, RAG (Retrieval-Augmented Generation), and LLM integration** (Ollama, Gemini free tier).

**Backend / API Developer** | *Project: Transport Management & Event Management Apps • Current, 1 Year*

- Developing **Python-based REST APIs** and backend business logic (FastAPI) for transport and event-management workflows.
- Building and supporting **WebSocket-based real-time features** for live application updates.
- Developed APIs with **multi-language support** to serve localized content across the application.
- Integrated **Google Maps APIs** for location tracking and distance calculation.
- Implemented **SMS notifications via MSG91** with DLT (regulatory) configuration for compliant delivery.
- Defined API contracts and coordinated backend integration with React / React Native frontend developers.
- Working on authentication, validation, database integration, application business rules, and debugging.
- Using Claude and other AI-assisted development tools in a structured workflow — personally reviewing architecture, business logic, generated code, and validation results before anything ships.

**Backend Developer** | *Project: Mariner — AI Course Assistant for Student Training*

- Worked on the backend of an **AI chat assistant** that answers students' questions about training courses, built on a **RAG pipeline with FAISS vector search and an OpenAI LLM**.
- Built backend services that **pull course details from external APIs and store them**, giving the retrieval layer clean, structured course data.
- Working knowledge of the RAG flow: embedding course content, FAISS similarity search, and passing retrieved context to the OpenAI model.

## PREVIOUS PROJECT EXPERIENCE (Compunet Connections)

**Food Delivery Application, Villa / Beach House Booking Platform, Animal Healthcare System, Labour & Delivery Management Application, Loan Application** — APIs, business logic, database operations, integrations, and application debugging.

## EDUCATION

**B.E., Computer Science & Engineering** — University College of Engineering, BIT Campus, Trichy | 2016–2020
**Higher Secondary (HSC)** — Dhanalakshmi Srinivasan Higher Secondary School | 2014–2016